PHASE 8: STATISTICAL VALIDATION REPORT

E-Commerce Capstone Project

1. Executive Summary

The Phase 8 notebook evaluates ten selected e-commerce hypotheses covering delivery performance, discounting and profitability, customer reviews, repeat purchasing, category sales, and customer loyalty. The Phase 3 framework establishes the broader capstone hypothesis structure; the Phase 8 notebook operationalizes a specific H1-H10 validation set. The report preserves those Phase 8 labels without renumbering.

Formal inferential testing is strongest for H1, H2, H3 and H10. H1 uses a one-sided two-proportion Z-test; H2 uses Welch's independent-samples T-test; H3 uses an independent-samples T-test; and H10 uses Welch's independent-samples T-test. H4-H9 are reported exactly as the notebook reports them: descriptive or correlation analyses without a formal hypothesis-test decision where no p-value is produced.

At the 5% significance level, H1 fails to reject H_0, H2 rejects H_0, H3 fails to reject H_0, and H10 rejects H_0. The clearest evidence is therefore around discount level versus profitability and loyalty versus customer revenue.

Hypothesis	Method	Statistic	P-value	Decision
H1 Delivery Delay → Cancellation	Two-Proportion Z-Test	-103.1500209	1.0	Fail to Reject H_0
H2 Discount → Profit	Welch Independent T-Test	89.9064979	0.0	Reject H_0
H3 Delivery Delay → Customer Satisfaction	Independent T-Test	0.9038300	0.3660893	Fail to Reject H_0
H4 Repeat Customers → Higher Revenue	Descriptive Comparison	—	—	Descriptive Analysis
H5 Product Category → Sales	Category-wise Sales	—	—	Descriptive Analysis
H6 Product Category → Profitability	Category-wise Profit	—	—	Descriptive Analysis
H7 Higher Discount → Lower Profit	Pearson Correlation	-0.5044666	—	Descriptive Analysis
H8 Review Score → Repeat Purchase	Group Comparison	—	—	Descriptive Analysis
H9 Delivery Performance → Review Score	Pearson Correlation	0.0029576	—	Descriptive Analysis
H10 Loyal Customers → Higher Revenue	Welch Independent T-Test	81.3509260	0.0	Reject H_0

2. Data, Preparation and Validation Approach

The notebook loads five cleaned datasets used directly in the Phase 8 analysis: Customers (90,000 × 5), Orders (99,999 × 13), Order Items (150,004 × 19), Products (32,951 × 16), and Reviews (97,019 × 7).

Dataset	Rows	Columns	Role
Customers	90,000	5	Customer identity, segment, age group, gender
Orders	99,999	13	Orders, status, payment, dates and delivery
Order Items	150,004	19	Quantity, price, discount, shipping, net amount
Products	32,951	16	Category, brand, cost, selling price, profit
Reviews	97,019	7	Review score, comments and review date

- Column names are standardized to lower-case, trimmed, and normalized.
- Duplicate records are removed before hypothesis analysis.
- Order status is normalized to lower-case and cancellation flags are created.
- Delivery group is defined as Delayed when delivery_delay > 0 and On Time otherwise.
- Hypothesis-specific datasets are created through joins on order_id, customer_id and product_id.
- Formal significance testing uses $\alpha = 0.05$.

Important limitation: several broader Phase 3 hypotheses require web-session, traffic, cart, checkout or product-page impression data that are not present in the Phase 8 notebook. Those hypotheses should not be represented as fully tested by this report.

3. Two Probability Assessments

The notebook first establishes two simple business probabilities used to contextualize the later hypothesis results: the overall cancellation probability and the repeat-customer probability.

Exact notebook stdout — Probability Assessments

```
Cancellation Probability: 5.05 %
Repeat Customer Probability: 50.44 %
```

Assessment	Notebook Result	Business Meaning
Cancellation Probability	5.05%	Approximately 5 out of every 100 distinct orders were cancelled.
Repeat Customer Probability	50.44%	Approximately half of customers with an order placed more than one order in the notebook's customer-order set.

H1. Delivery Delay → Cancellation

Business Question: Are delayed orders associated with a higher cancellation rate than on-time orders?

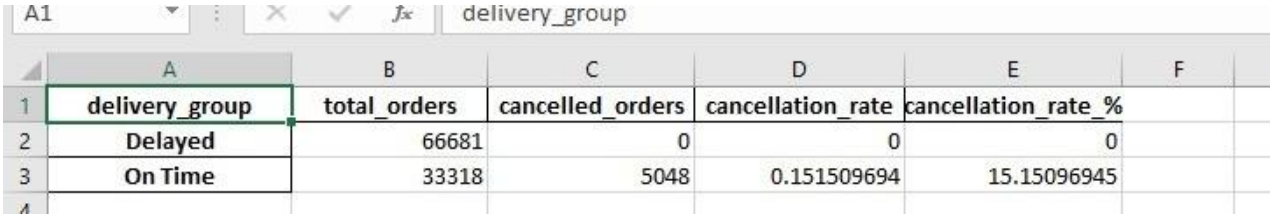
H_0: There is no significant difference in cancellation rates between delayed and on-time orders.

H_1: Delayed orders have a higher cancellation rate.

Method: Two-Proportion Z-Test; one-sided alternative='larger'.

Hypothesis Validation Approach: The notebook compares cancellation proportions after grouping orders into Delayed and On Time.

Visual Output



	A	B	C	D	E	F
1	delivery_group	total_orders	cancelled_orders	cancellation_rate	cancellation_rate_%	
2	Delayed	66681	0	0	0	
3	On Time	33318	5048	0.151509694	15.15096945	

Exact notebook stdout – H1 Delivery Group Summary

```
               total_orders  cancelled_orders  cancellation_rate  delivery_group

Delayed               66681                 0            0.00000
On Time               33318              5048            0.15151

               cancellation_rate_%  delivery_group
Delayed                   0.000000
On Time                  15.150969
```

Exact notebook stdout – H1 Z-Test

```
Z-statistic: -103.15002090572847
P-value: 1.0
Decision: Fail to Reject H0
```

Statistical Decision

Decision: Fail to Reject H_0 at $\alpha = 0.05$.

Interpretation: Delayed orders show 0 cancelled orders out of 66,681 (0.00000%), while On Time shows 5,048 cancellations out of 33,318 (15.150969%). The one-sided test returns z = -103.15002090572847 and p = 1.0, so the directional claim that delayed orders cancel more is not supported. This result is strongly affected by how delivery_group is constructed and should be validated before operational use.

Business Recommendation: Validate delivery-delay values and the treatment of cancelled or missing-delivery records before using delay as a causal cancellation driver.

H2. Discount → Profit

Business Question: Does profit differ significantly between low/moderate-discount and high-discount groups?

H_0: There is no significant difference in mean profit between the two discount groups.

H_1: Mean profit differs between low/moderate and high discount groups.

Method: Welch's Independent T-Test.

Hypothesis Validation Approach: Profit is calculated from net_product_amount minus product cost; observations are split at a 20% discount threshold.

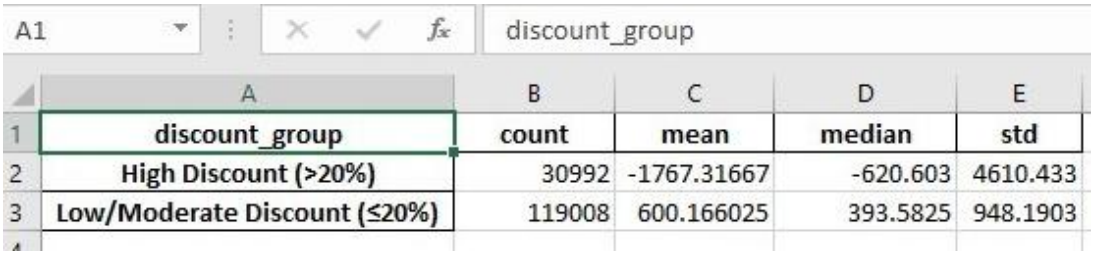
Visual Output

Exact notebook stdout — H2 Discount Group Summary

```
                              count         mean    median          std
discount_group
High Discount (>20%)          30992 -1767.316667 -620.6030  4610.433405
Low/Moderate Discount (≤20%)  119008   600.166025  393.5825   948.190267
```

Exact notebook stdout — H2 Welch T-Test

```
T-statistic: 89.90649787811803
P-value: 0.0
Decision: Reject H0
```



A1 | discount_group

	A	B	C	D	E
1	discount_group	count	mean	median	std
2	High Discount (>20%)	30992	-1767.31667	-620.603	4610.433
3	Low/Moderate Discount (≤20%)	119008	600.166025	393.5825	948.1903

Statistical Decision

Decision: Reject H_0 at $\alpha = 0.05$.

Interpretation: High-discount items (>20%) have mean profit of -1,767.316667 versus 600.166025 for the Low/Moderate group. The Welch test returns t = 89.90649787811803 and p = 0.0 in the notebook. The difference is statistically significant and economically material.

Business Recommendation: Introduce category-specific discount ceilings and evaluate promotions using contribution margin, not sales lift alone.

H3. Delivery Delay → Customer Satisfaction

Business Question: Does average review score differ between delayed and on-time orders?

H_0: Average review score is the same for delayed and on-time orders.

H_1: Average review score differs between delayed and on-time orders.

Method: Independent T-Test (Welch implementation with equal_var=False).

Hypothesis Validation Approach: Orders are joined to review scores and the mean, median and standard deviation are compared by delivery group.

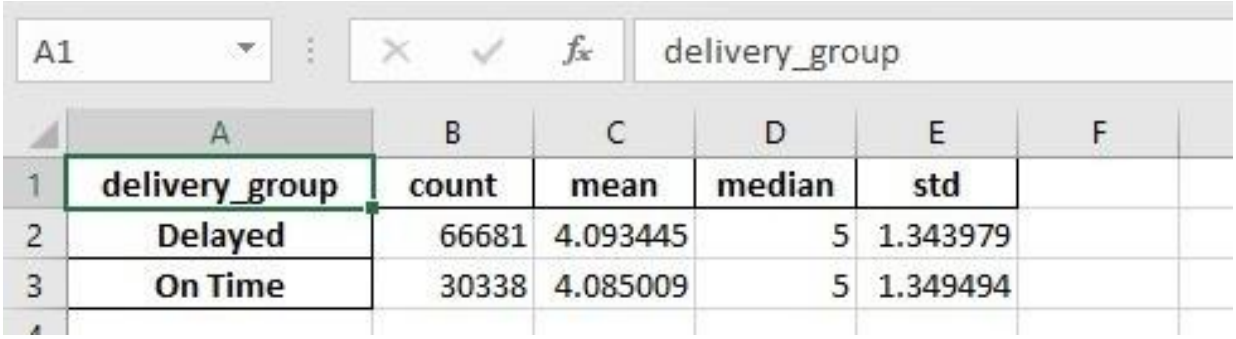
Visual Output

Exact notebook stdout — H3 Review Score by Delivery Group

```
                count      mean  median       std
delivery_group
Delayed         66681  4.093445     5.0  1.343979
On Time         30338  4.085009     5.0  1.349494
```

Exact notebook stdout — H3 Independent T-Test

```
T-statistic: 0.9038299790689821
P-value: 0.3660892858513811
```



	A	B	C	D	E	F
1	delivery_group	count	mean	median	std	
2	Delayed	66681	4.093445	5	1.343979	
3	On Time	30338	4.085009	5	1.349494	

Statistical Decision

Decision: Fail to Reject H_0 at $\alpha = 0.05$.

Interpretation: Delayed orders have mean review 4.093445 versus 4.085009 for On Time. The test returns t = 0.9038299790689821 and p = 0.3660892858513811. The evidence is insufficient to conclude a statistically significant difference in average review score.

Business Recommendation: Do not treat delivery delay as a standalone satisfaction predictor; combine delivery metrics with product, price, returns and review-content factors.

H4. Repeat Customers → Higher Revenue

Business Question: Do repeat customers generate higher revenue than one-time customers?

H_0: Repeat and one-time customers have the same revenue distribution.

H_1: Repeat customers generate higher revenue.

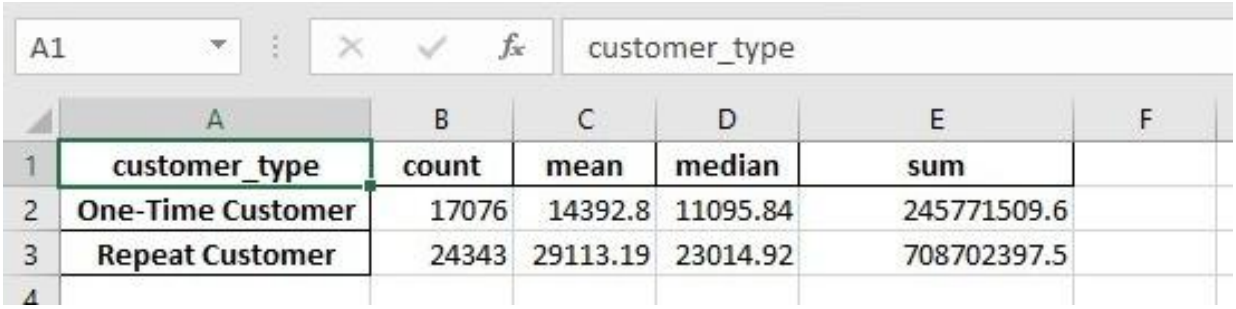
Method: Descriptive comparison; no formal p-value reported in the notebook.

Hypothesis Validation Approach: Customer revenue is summed from order-item net product amount and customers are classified by whether their order count exceeds one.

Visual Output

Exact notebook stdout — H4 Revenue by Customer Type

```
                    count          mean     median           sum
customer_type
One-Time Customer  17076  14392.803325  11095.835  2.457715e+08
Repeat Customer    24343  29113.190549  23014.919  7.087024e+08
```



A1 | customer_type

	A	B	C	D	E	F
1	customer_type	count	mean	median	sum	
2	One-Time Customer	17076	14392.8	11095.84	245771509.6	
3	Repeat Customer	24343	29113.19	23014.92	708702397.5	
4						

Statistical Decision

Decision: Descriptive Analysis; no formal reject/fail-to-reject decision.

Interpretation: Repeat customers show mean revenue of 29,113.190549 versus 14,392.803325 for one-time customers, with total revenue of about 708.7M versus 245.8M. The descriptive difference is large, but the notebook does not provide an inferential test for H4.

Business Recommendation: Build retention journeys around the second purchase and then add a formal test in a later validation phase.

H5. Product Category → Sales

Business Question: Are sales concentrated in a limited number of product categories?

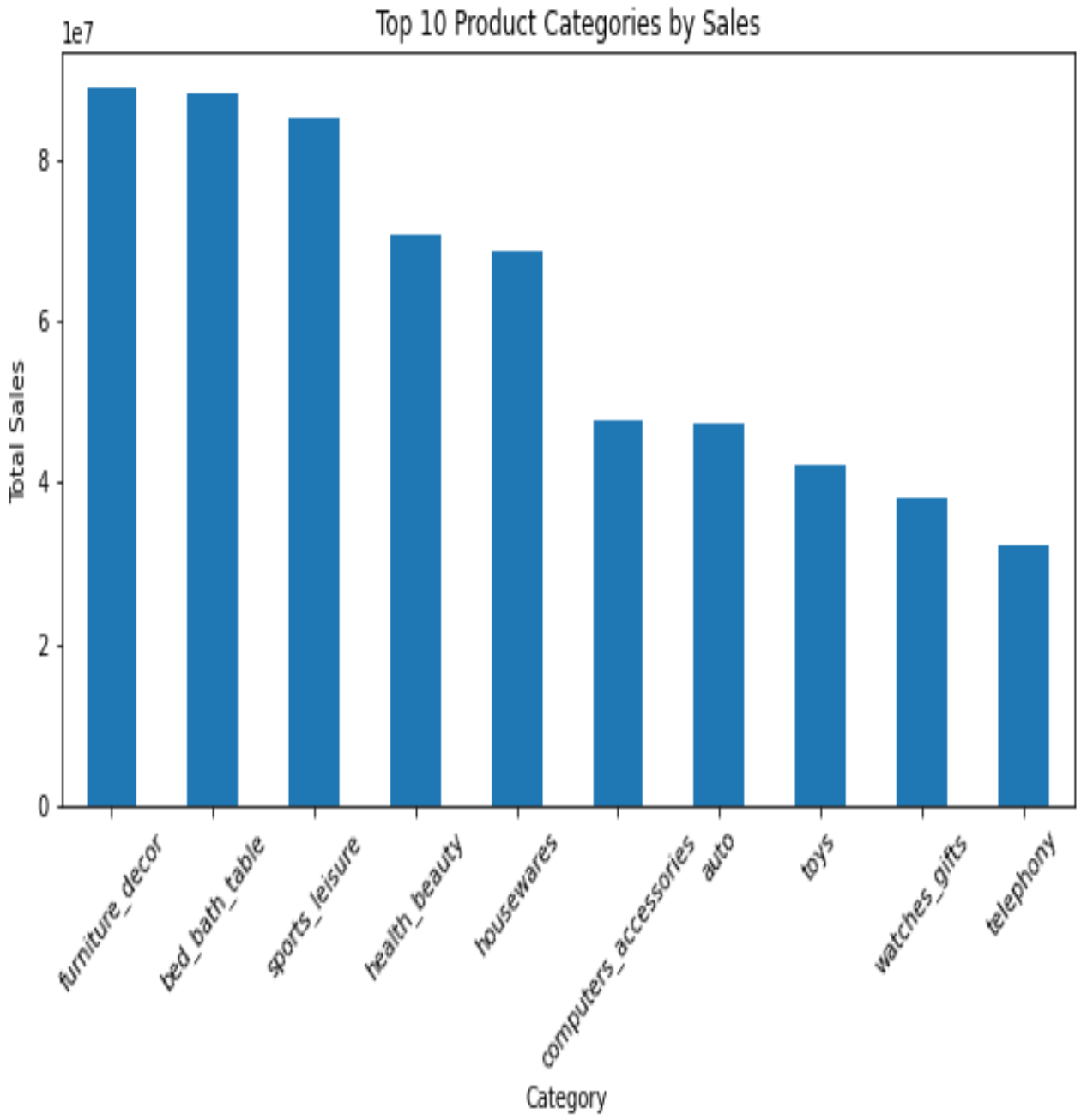
H_0: Sales are not materially differentiated by product category.

H_1: A limited number of categories generate disproportionately high sales.

Method: Category-wise Sales Analysis; descriptive ranking.

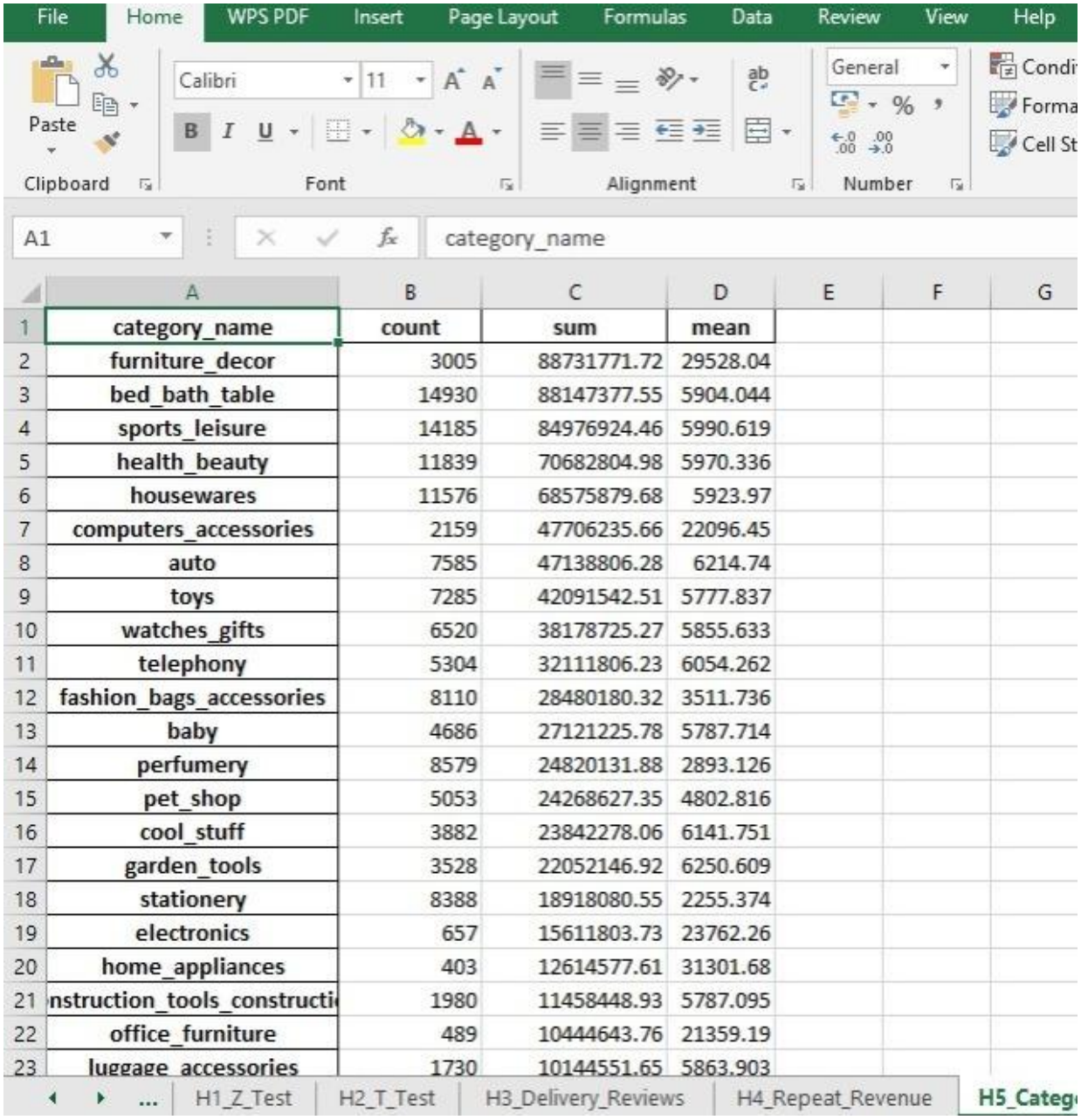
Hypothesis Validation Approach: Net product amount is aggregated by category and the top ten categories are ranked by total sales.

Visual Output



Exact notebook stdout — H5 Top 10 Categories by Sales

category_name	count	sum	mean
furniture_decor	3005	8.873177e+07	29528.043834
bed_bath_table	14930	8.814738e+07	5904.044042
sports_leisure	14185	8.497692e+07	5990.618573
health_beauty	11839	7.068280e+07	5970.335753
housewares	11576	6.857588e+07	5923.970255
computers_accessories	2159	4.770624e+07	22096.450050
auto	7585	4.713881e+07	6214.740446
toys	7285	4.209154e+07	5777.836995
watches_gifts	6520	3.817873e+07	5855.632710
telephony	5304	3.211181e+07	6054.262109



	A	B	C	D
1	category_name	count	sum	mean
2	furniture_decor	3005	88731771.72	29528.04
3	bed_bath_table	14930	88147377.55	5904.044
4	sports_leisure	14185	84976924.46	5990.619
5	health_beauty	11839	70682804.98	5970.336
6	housewares	11576	68575879.68	5923.97
7	computers_accessories	2159	47706235.66	22096.45
8	auto	7585	47138806.28	6214.74
9	toys	7285	42091542.51	5777.837
10	watches_gifts	6520	38178725.27	5855.633
11	telephony	5304	32111806.23	6054.262
12	fashion_bags_accessories	8110	28480180.32	3511.736
13	baby	4686	27121225.78	5787.714
14	perfumery	8579	24820131.88	2893.126
15	pet_shop	5053	24268627.35	4802.816
16	cool_stuff	3882	23842278.06	6141.751
17	garden_tools	3528	22052146.92	6250.609
18	stationery	8388	18918080.55	2255.374
19	electronics	657	15611803.73	23762.26
20	home_appliances	403	12614577.61	31301.68
21	nstruction_tools_constructi	1980	11458448.93	5787.095
22	office_furniture	489	10444643.76	21359.19
23	luggage_accessories	1730	10144551.65	5863.903

H1_Z_Test | H2_T_Test | H3_Delivery_Reviews | H4_Repeat_Revenue | H5_Categ

Statistical Decision

Decision: Descriptive Analysis; no formal significance test reported.

Interpretation: Furniture_decor leads at approximately 88.73177M, narrowly ahead of bed_bath_table at approximately 88.14738M, with sports_leisure at approximately 84.97692M. The result supports strong category differentiation, but the notebook does not quantify concentration statistically.

Business Recommendation: Prioritize leading categories for assortment, inventory and marketing while using the long tail for targeted optimization.

H6. Product Category → Profitability

Business Question: Does profitability vary substantially across product categories?

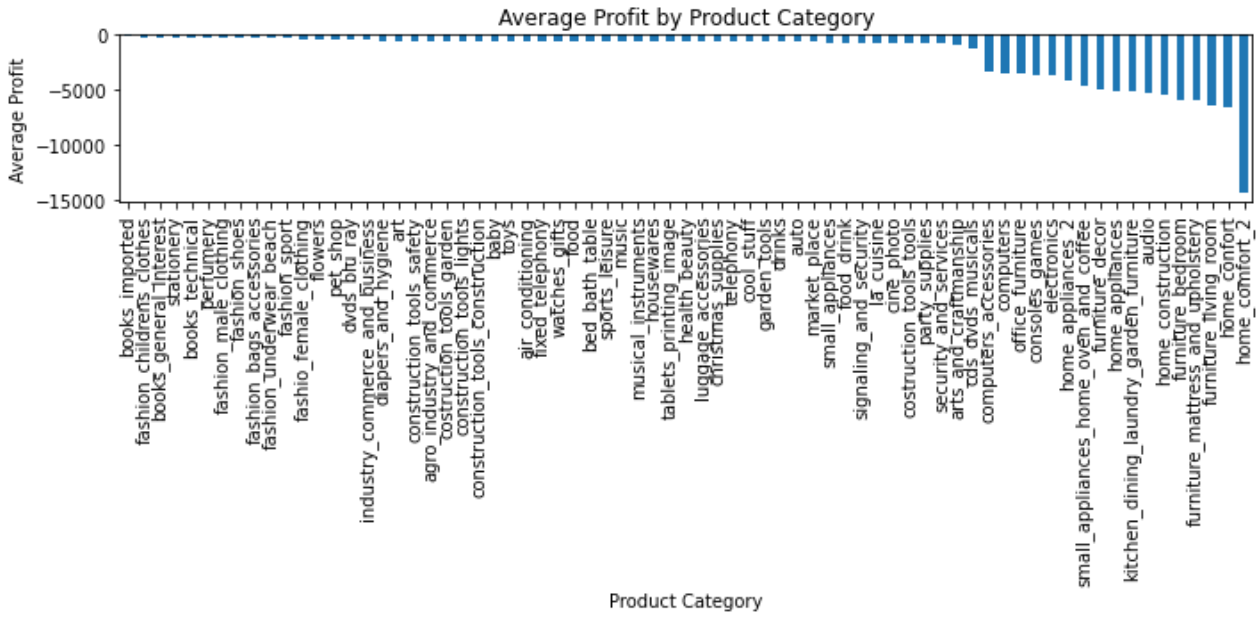
H_0: Average profitability is the same across product categories.

H_1: Profitability differs across product categories.

Method: Category-wise Profit Analysis; descriptive comparison.

Hypothesis Validation Approach: Profit is calculated for order items and summarized by category using count, mean, median and sum.

Visual Output

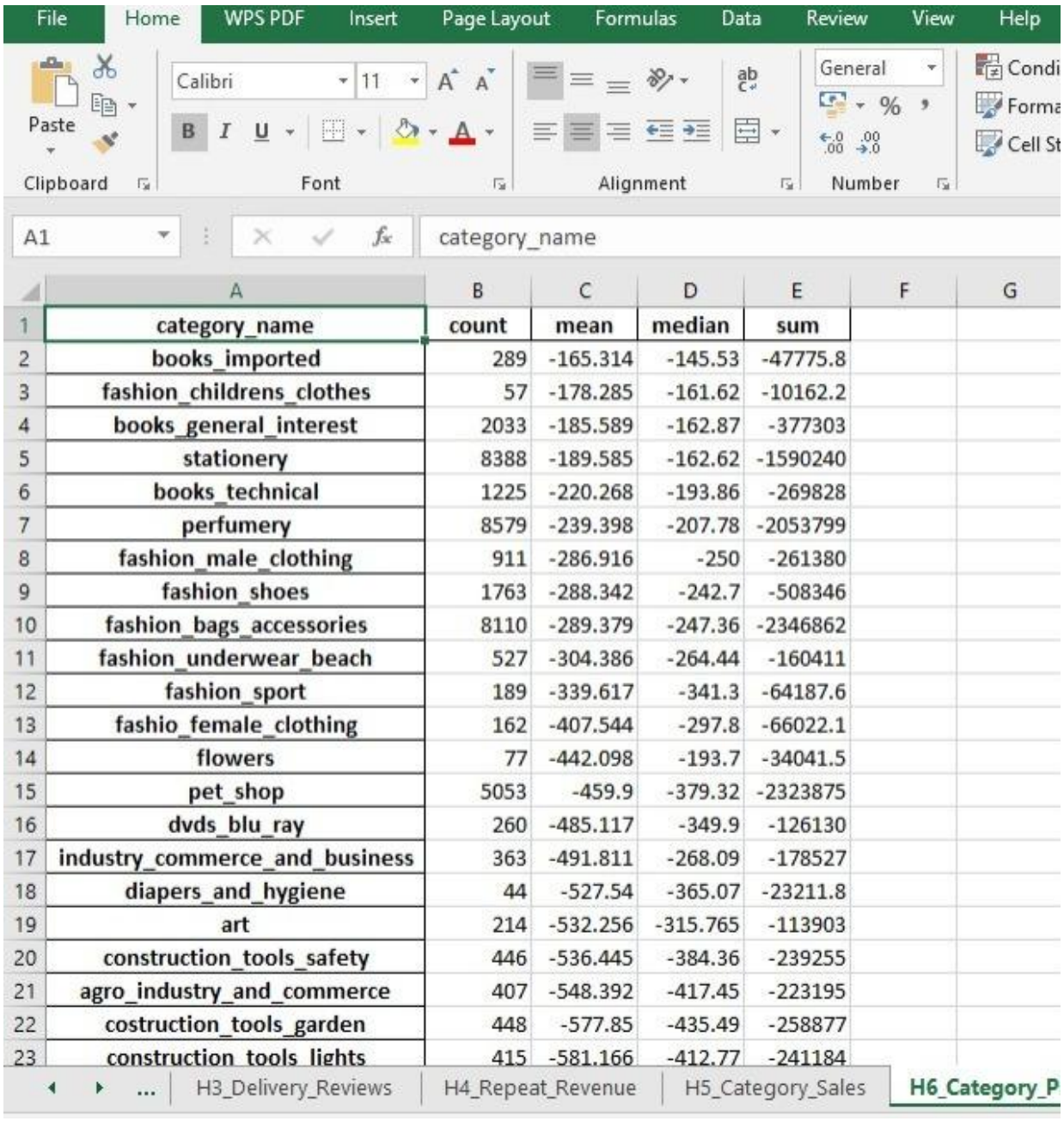


Exact notebook stdout — H6 Profitability Output (top/bottom rows)

```
                                 count          mean     median         sum
category_name
books_imported                     289   -165.314083   -145.530   -47775.77
fashion_childrens_clothes           57   -178.284737   -161.620   -10162.23
books_general_interest            2033   -185.589061   -162.870 -377302.56
stationery                        8388   -189.585135   -162.620 -1590240.11
books_technical                   1225   -220.267510   -193.860 -269827.70
...                                ...           ...        ...         ...
furniture_bedroom                   39  -5860.596923  -4390.140 -228563.28
furniture_mattress_and_upholstery   13  -5934.949231  -1921.280  -77154.34
furniture_living_room              110  -6431.021364  -4284.830 -707412.35
home_confort                       110  -6613.418727  -2953.005 -727476.06
home_comfort_2                       5 -14414.252000 -16827.360  -72071.26

[71 rows x 4 columns]
```

..



	A	B	C	D	E	F	G
1	category_name	count	mean	median	sum		
2	books_imported	289	-165.314	-145.53	-47775.8		
3	fashion_childrens_clothes	57	-178.285	-161.62	-10162.2		
4	books_general_interest	2033	-185.589	-162.87	-377303		
5	stationery	8388	-189.585	-162.62	-1590240		
6	books_technical	1225	-220.268	-193.86	-269828		
7	perfumery	8579	-239.398	-207.78	-2053799		
8	fashion_male_clothing	911	-286.916	-250	-261380		
9	fashion_shoes	1763	-288.342	-242.7	-508346		
10	fashion_bags_accessories	8110	-289.379	-247.36	-2346862		
11	fashion_underwear_beach	527	-304.386	-264.44	-160411		
12	fashion_sport	189	-339.617	-341.3	-64187.6		
13	fashio_female_clothing	162	-407.544	-297.8	-66022.1		
14	flowers	77	-442.098	-193.7	-34041.5		
15	pet_shop	5053	-459.9	-379.32	-2323875		
16	dvds_blu_ray	260	-485.117	-349.9	-126130		
17	industry_commerce_and_business	363	-491.811	-268.09	-178527		
18	diapers_and_hygiene	44	-527.54	-365.07	-23211.8		
19	art	214	-532.256	-315.765	-113903		
20	construction_tools_safety	446	-536.445	-384.36	-239255		
21	agro_industry_and_commerce	407	-548.392	-417.45	-223195		
22	costruction_tools_garden	448	-577.85	-435.49	-258877		
23	construction_tools_lights	415	-581.166	-412.77	-241184		

Statistical Decision

Decision: Descriptive Analysis; no formal multi-group significance test reported.

Interpretation: Profitability varies widely across the 71 categories. The printed output ranges from books_imported at -165.314083 average profit to home_comfort_2 at -14,414.252000. The plotted values are negative across the displayed categories. These differences are economically important but not statistically tested in the notebook.

Business Recommendation: Audit cost structure, pricing, shipping and discounting by category before scaling negative-margin assortments.

H7. Higher Discount → Lower Profit

Business Question: Is higher discount percentage associated with lower profit?

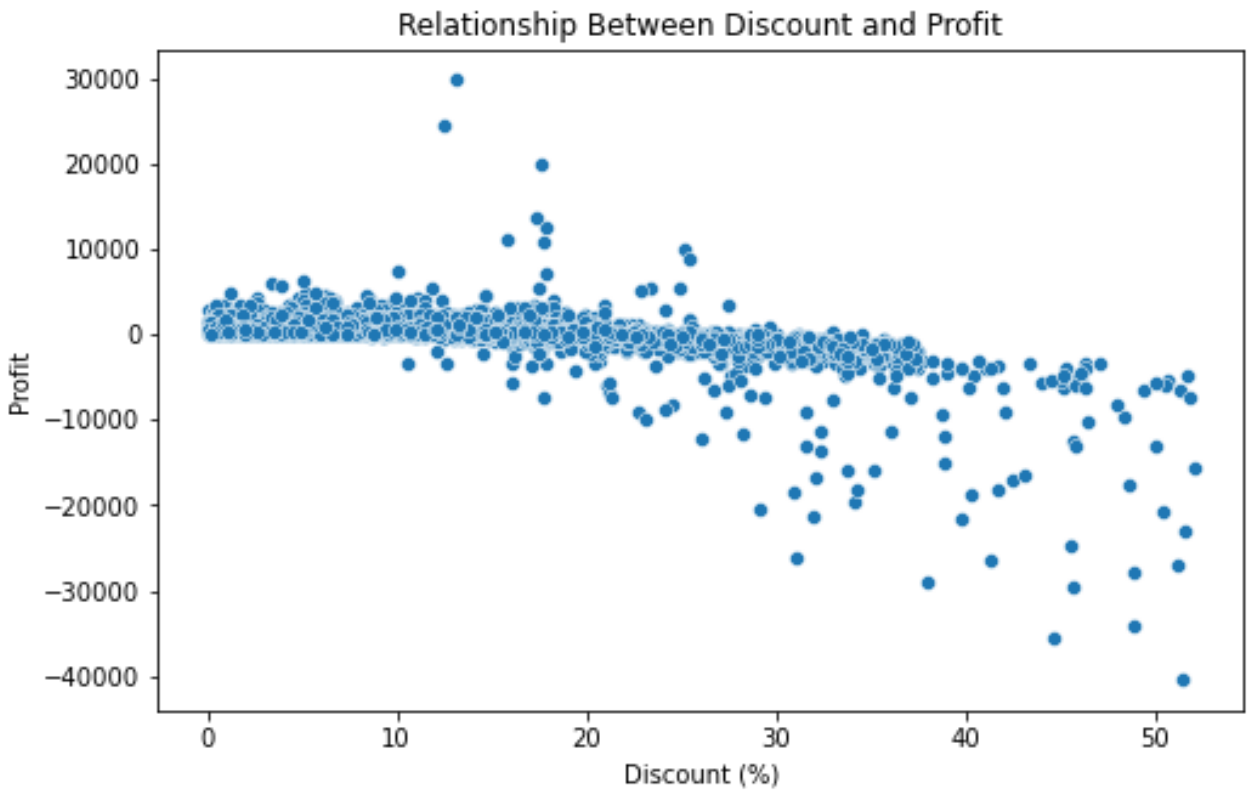
H_0: Discount percentage has no linear relationship with profit.

H_1: Higher discounts are associated with lower profit.

Method: Pearson Correlation Analysis.

Hypothesis Validation Approach: The notebook calculates the correlation between discount percentage and profit after dropping missing values.

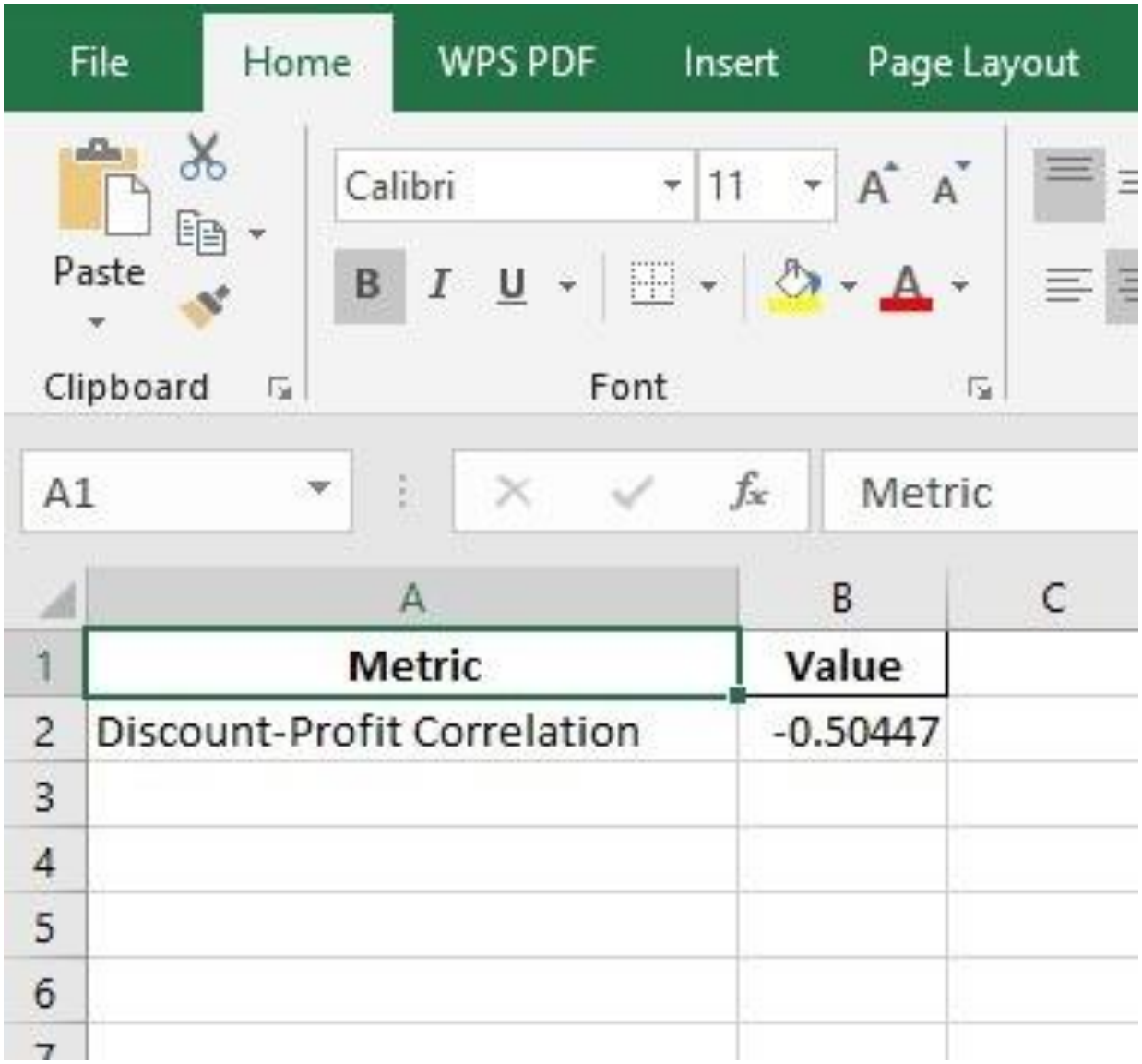
Visual Output



Exact notebook stdout — H7 Correlation / Interpretation

```
Discount-Profit Correlation: -0.5044666006397452
Higher discounts are associated with lower profit.
```

Exact notebook stdout: H7 correlation and interpretation.



Statistical Decision

Decision: Descriptive Analysis; no p-value reported.

Interpretation: The correlation is -0.5044666006397452, indicating a moderate negative association: as discount percentage increases, profit tends to decline. The notebook also explicitly prints that higher discounts are associated with lower profit.

Business Recommendation: Use discount guardrails and protect contribution margin, especially for high-volume promotional campaigns.

H8. Review Score → Repeat Purchase

Business Question: Are higher review scores associated with repeat purchasing?

H_0: Repeat and non-repeat customers have the same review-score distribution.

H_1: Higher review scores are associated with repeat purchasing.

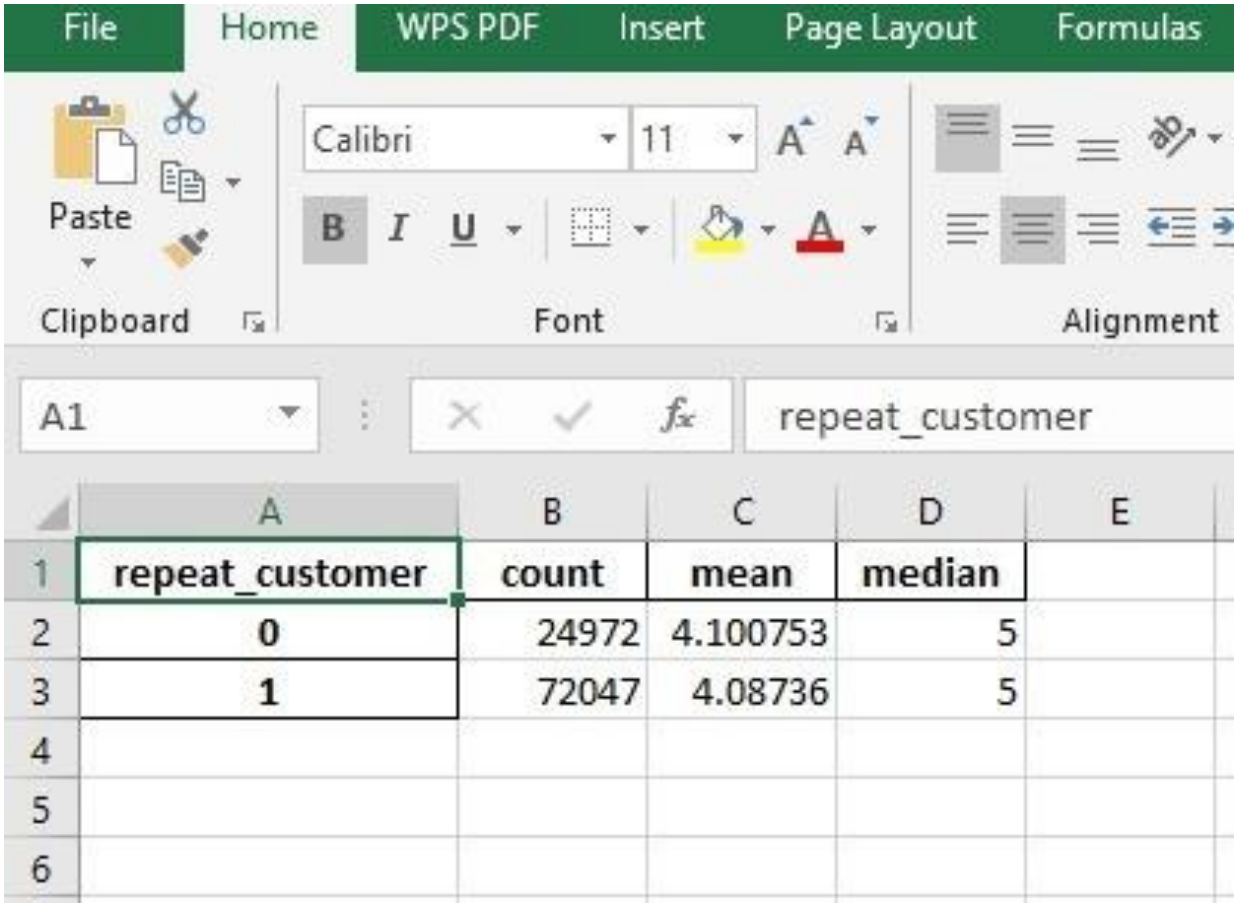
Method: Group Comparison; descriptive comparison of review scores by repeat-customer status.

Hypothesis Validation Approach: Reviews are linked to customers through order_id, then review scores are compared between repeat and non-repeat customers.

Exact Visual Output

Exact notebook stdout — H8 Review Score by Repeat Status

```
                 count      mean  median
repeat_customer
0                24972  4.100753     5.0
1                72047  4.087360     5.0
```



	A	B	C	D	E
1	repeat_customer	count	mean	median	
2	0	24972	4.100753	5	
3	1	72047	4.08736	5	
4					
5					
6					

Statistical Decision

Decision: Descriptive Analysis; no formal p-value reported.

Interpretation: Non-repeat customers have a slightly higher mean review score (4.100753) than repeat customers (4.087360), while both medians are 5.0. The observed direction is opposite to the hypothesis, so the notebook does not support the proposed positive relationship.

Business Recommendation: Do not rely on review score alone as a retention lever; combine review content, product value, service experience and purchase timing.

H9. Delivery Performance → Review Score

Business Question: Is delivery delay linearly related to customer review score?

H_0: Delivery delay has no relationship with review score.

H_1: Delivery delay is associated with review score.

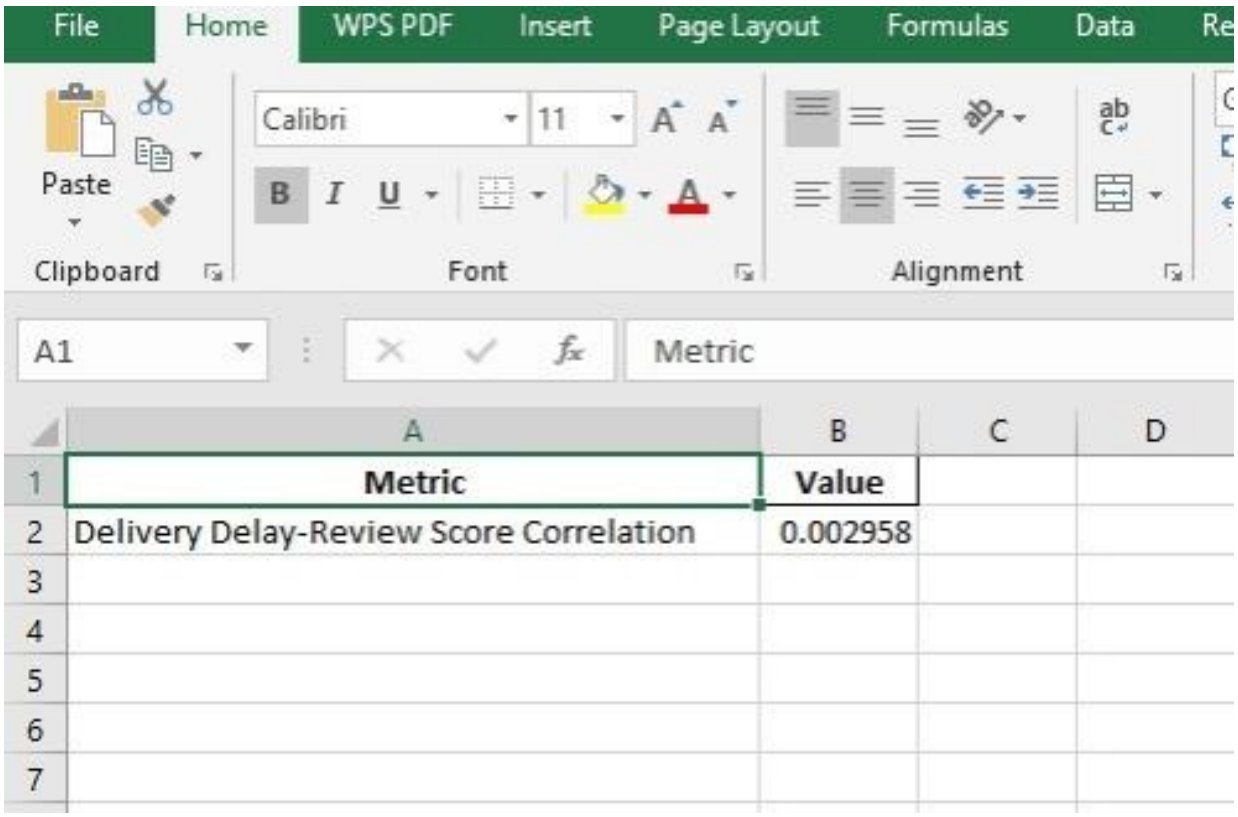
Method: Pearson Correlation Analysis.

Hypothesis Validation Approach: The notebook drops missing delay/review values and computes the Pearson correlation.

Exact Visual Output

Exact notebook stdout — H9 Delivery Delay / Review Score

```
Delivery Delay-Review Score Correlation: 0.0029575599333364713
```



Statistical Decision

Decision: Descriptive Analysis; no p-value reported.

Interpretation: The correlation is 0.0029575599333364713, which is effectively zero. This indicates no meaningful linear relationship between delivery delay and review score in the analysed sample.

Business Recommendation: Treat delivery delay as one operational KPI among several rather than as the primary explanation for rating outcomes.

H10. Loyal Customers → Higher Revenue

Business Question: Do loyal customers generate significantly higher revenue than non-loyal customers?

H_0: Loyal and non-loyal customers generate the same revenue.

H_1: Loyal customers generate higher average revenue.

Method: Welch's Independent T-Test.

Hypothesis Validation Approach: Customers are classified as Loyal when they have more than one distinct order; customer revenue is then compared between loyalty groups.

Exact Visual Output

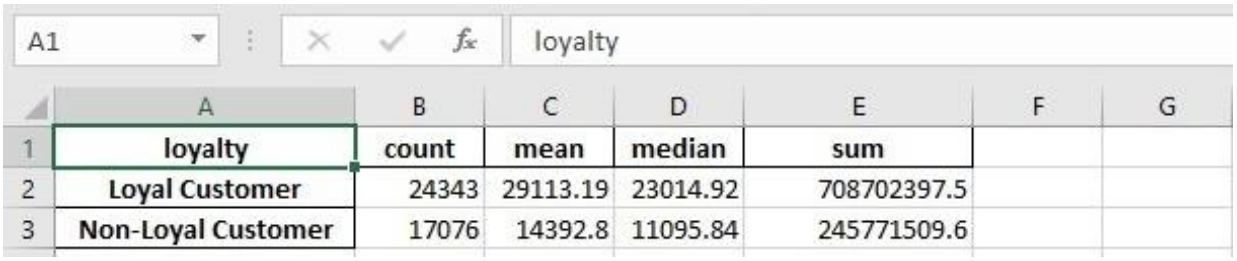
Exact notebook stdout — H10 Loyalty Revenue Summary

```
                    count          mean     median           sum
loyalty
Loyal Customer      24343  29113.190549  23014.919  7.087024e+08
Non-Loyal Customer  17076  14392.803325  11095.835  2.457715e+08
```

Exact notebook stdout: H10 loyalty-group revenue summary.

Exact notebook stdout — H10 Welch T-Test

```
T-statistic: 81.35092602593404
P-value: 0.0
Decision: Reject H0
```



Statistical Decision

Decision: Reject H_0 at $\alpha = 0.05$.

Interpretation: Loyal customers have mean revenue of 29,113.190549 versus 14,392.803325 for non-loyal customers. The Welch test returns t = 81.35092602593404 and p = 0.0 in the notebook. This is strong evidence of a difference in average revenue between the two groups.

Business Recommendation: Prioritize retention programs, second-purchase journeys and loyalty incentives while measuring incremental customer lifetime value.

4. Final Validation Table

The table below consolidates the complete H1-H10 decision layer. Formal statistical decisions are shown only where the Phase 8 notebook produced a formal test statistic and p-value. Descriptive hypotheses are intentionally marked as descriptive rather than being given an invented significance decision.

Hypothesis	Test / Method	Test Statistic	Decision / Status
H1 Delivery Delay → Cancellation	Two-Proportion Z-Test	-103.1500209	Fail to Reject H_0
H2 Discount → Profit	Welch Independent T-Test	89.9064979	Reject H_0
H3 Delivery Delay → Customer Satisfaction	Independent T-Test	0.9038300	Fail to Reject H_0
H4 Repeat Customers → Higher Revenue	Descriptive Comparison	—	Descriptive Analysis
H5 Product Category → Sales	Category-wise Sales	—	Descriptive Analysis
H6 Product Category → Profitability	Category-wise Profit	—	Descriptive Analysis
H7 Higher Discount → Lower Profit	Pearson Correlation	-0.5044666	Descriptive Analysis
H8 Review Score → Repeat Purchase	Group Comparison	—	Descriptive Analysis
H9 Delivery Performance → Review Score	Pearson Correlation	0.0029576	Descriptive Analysis
H10 Loyal Customers → Higher Revenue	Welch Independent T-Test	81.3509260	Reject H_0

	A	B	C	D	E	F
1	Hypothesis	Test	Test Statistic	P-value	Alpha	Decision
2	H1: Delivery Delay → Cancellation	Two-Proportion Z-Test	-103.1500209	1	0.05	Fail to Reject H0
3	H2: Discount → Profit	Welch's Independent T-Test	89.90649788	0	0.05	Reject H0
4						
5						

5. Probability and Testing Flow Summary

Stage	Output
10 Hypotheses	H1-H10 evaluated in sequence.
Probability Assessment 1	Cancellation Probability = 5.05%.
Probability Assessment 2	Repeat Customer Probability = 50.44%.
H1	Two-Proportion Z-Test → z = -103.1500, p = 1.0 → Fail to Reject H_0.
H2	Welch T-Test → t = 89.9065, p = 0.0 → Reject H_0.
Additional formal checks	H3 T-Test and H10 Welch T-Test are also reported by the notebook.
Remaining hypotheses	H4-H9 retained as descriptive/correlation analyses where no formal p-value is reported.
Final output	One consolidated H1-H10 validation table.

6. Cross-Hypothesis Findings

- The strongest statistical evidence is concentrated in economics and loyalty: H2 shows a significant profit difference by discount group, and H10 shows a significant revenue difference between loyal and non-loyal customers.
- Delivery results are consistently weak as standalone explanatory factors: H1 does not support higher cancellation in delayed orders, H3 does not show a significant review-score difference, and H9 has an almost zero correlation.
- H4 supports a strong descriptive revenue gap between repeat and one-time customers, but it needs a formal inferential test in a future validation phase.
- H5 and H6 show substantial category-level differences in sales and profitability, but the notebook does not perform a formal multi-group significance test.
- H7 provides a moderate negative discount-profit relationship, consistent with the significant group difference observed in H2.
- H8 does not support the proposed direction: repeat customers have a slightly lower mean review score than non-repeat customers.

7. Methodological Limitations and Validation Notes

- The Phase 8 notebook uses delivery_group = Delayed when delivery_delay > 0 and On Time otherwise. Missing or non-positive delays may therefore be classified as On Time.
- The H1 cancellation comparison is sensitive to that delivery-group construction and should be validated against the raw delivery-date fields.
- The notebook's p-value of 0.0 for H2 and H10 should be interpreted as an extremely small p-value as printed by the software, not as a literal mathematical zero.
- H4-H9 do not have formal inferential p-values in the notebook, so their results are descriptive and should not be reported as statistically significant.
- H10 loyalty is defined as more than one order. This is a practical repeat-customer definition, but it is not a complete fixed-window CLV or churn framework.
- Several wider Phase 3 hypotheses require session, traffic, cart, checkout, promotion-type or product-page data that are not present in the Phase 8 notebook.
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8. Business Recommendations

- Set discount guardrails by category and enforce minimum contribution-margin thresholds for high-discount campaigns.
- Build a second-purchase conversion program for one-time customers using personalized follow-ups, cross-sell offers and loyalty incentives.
- Protect and scale high-revenue categories while auditing negative-margin categories for pricing, sourcing, shipping and discount causes.
- Do not use delivery delay alone as a predictor of cancellation or satisfaction until data-quality and cancellation-timing issues are resolved.
- Add formal tests for the descriptive hypotheses in a later phase and extend the data model with traffic/session and funnel events where conversion hypotheses need to be tested.

9. Conclusion

The Phase 8 statistical validation provides its clearest evidence around discount discipline and customer loyalty. The high-discount group shows a statistically significant reduction in profit relative to the low/moderate group, while loyal customers show a statistically significant higher mean revenue than non-loyal customers. By contrast, the current data does not provide statistical support for delivery delay as a standalone driver of cancellation or customer satisfaction. Category-level sales and profitability remain materially differentiated, but those relationships are presented as descriptive because the notebook does not apply formal multi-group significance testing.

Overall, the correct management response is to protect margin, strengthen repeat purchase and loyalty, prioritize profitable categories, and expand the dataset before making stronger claims about conversion, causal delivery effects or customer churn.